



General Chemistry I

Recitation 11

Midterm II Review

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Dec. 8st

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		
															Pnictogens	Chalcogens	Halogens			
1	1 H Hydrogen 1.008	Atomic Symbol Name Weight C Solid Hg Liquid H Gas Rf Unknown Metals Alkali metals Alkaline earth metals Lanthanoids Actinoids Transition metals Post-transition metals Metalloids Nonmetals Reactive nonmetals Noble gases																2 He Helium 4.0026		
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122													5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305													13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798		
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29		
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71 89–103	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)		
7	87 Fr Francium (223)	88 Ra Radium (226)		104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)		
For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.																				
			6 7	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97		
				89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)		

Quiz

1. Write balanced equations for the following reactions. Use structural formulas to represent the organic compounds.

(a) The complete combustion (to CO_2 and H_2O) of cyclopropanol

(b) The dehydration of ethanol to give ethene

2. How many possible isomers correspond to $\text{C}_2\text{H}_2\text{BrCl}$? Remember that a simple rotation of an entire molecule does not give a different isomer. The molecule contains a central $\text{C}=\text{C}$ bond.

Homework5

1. Describe the bonding in the bent molecule OF₂. Predict its energy level diagram and electron configuration.

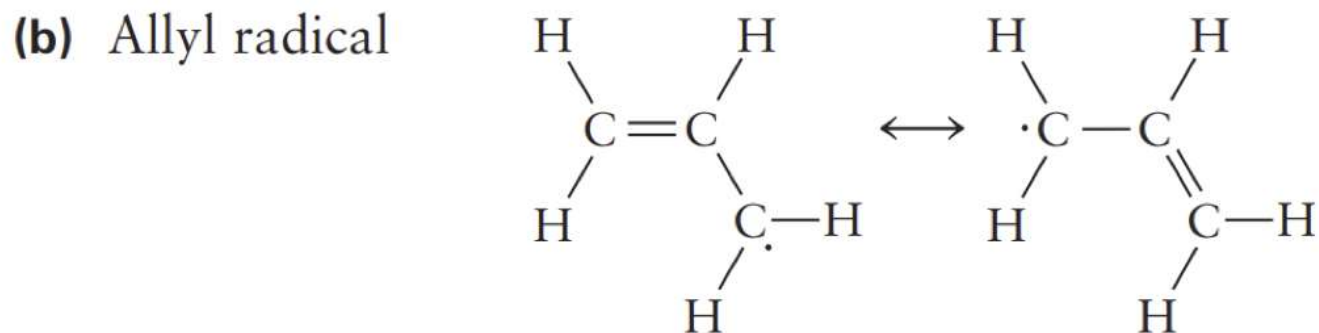
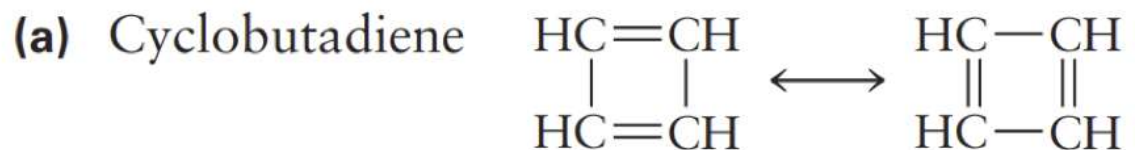
The OF₂ molecule has 20 valence electrons. Two O—F σ bonds form from overlap of sp^2 hybrid orbitals on the central O atom with $2p$ orbitals on the two F atoms. These two bonds use four electrons. Ten electrons occupy orbitals that are not properly oriented for overlap with other atoms' orbitals: two in the $2s$ orbital on F1; two in a $2s$ orbital on F2; two in the $2p$ orbital on F1, two in a $2p$ orbital on F2; two in the third sp^2 hybrid orbital on the O. This leaves six electrons and three p orbitals (one each on three atoms). These p orbitals overlap to form the π molecular orbital system shown in Text Figure 6.47. The resulting MO's accommodate electrons as follows: $(\pi)^2(\pi^{nb})^2(\pi^*)^2$. The bond order of the molecule based on just the σ bonding is 2. The π system adds nothing to this. For this reason the bonding in OF₂ can also be described as the result of σ overlap of sp^3 hybrid orbitals on the central O atom with $2p$ orbitals on the two F atoms and the assignment of the other 16 electrons to lone pairs (three of them on each F and two on the O).

Homework5

2. Discuss the nature of the bonding in the nitrate ion (NO_3^-). Draw the possible Lewis resonance diagrams for this ion. Use the VSEPR theory to determine the steric number, the hybridization of the central N atom, and the geometry of the ion. Show how the use of resonance structures can be avoided by introducing a de-localized π MO. What bond order is predicted by the MO model for the N-O bonds in the nitrate ion?

Homework5

3. For each of the following molecules, construct the π MOs from the $2p_z$ atomic orbitals perpendicular to the plane of the carbon atoms.



Indicate which, if any, of these orbitals have identical energies from symmetry considerations. Show the number of electrons occupying each π MO in the ground state, and indicate whether either or both of the molecules are paramagnetic. Assume that the C atoms in the allyl radical are all sp^2 hybridized.

MO and molecular structure

MO theory

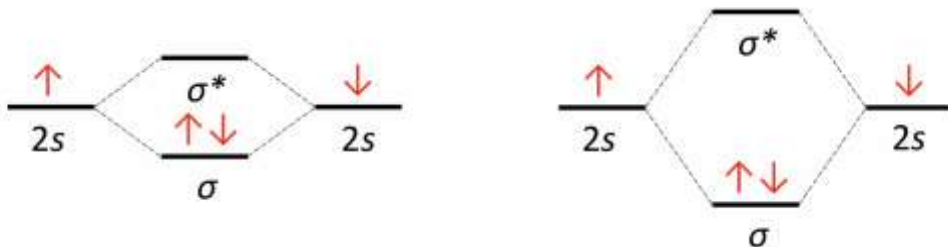
Molecular orbitals arise from allowed interactions between atomic orbitals (Atomic orbital overlap)

How to form efficient overlap:

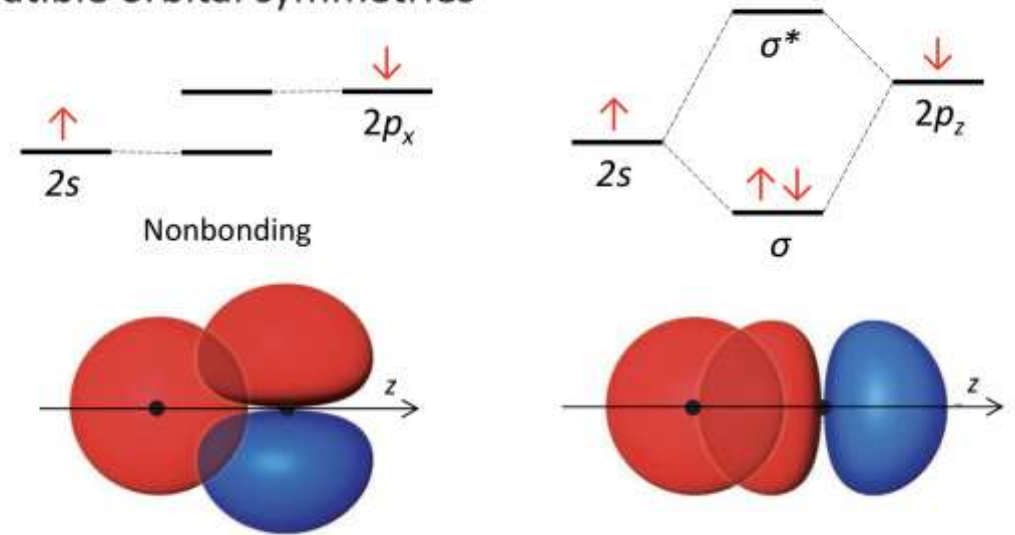
- symmetries are compatible with each other.
- atomic orbitals are close in energy.
- the number of molecular orbitals formed must be equal to the number of atomic orbitals in the atoms being combined to form the molecule.



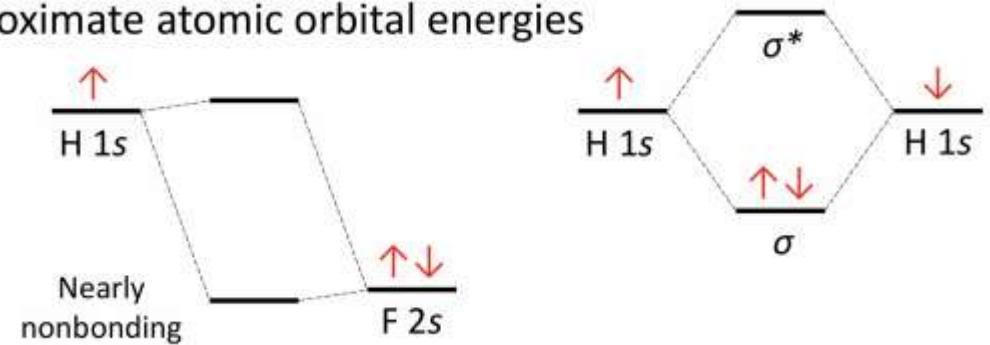
① Maximum orbital overlap



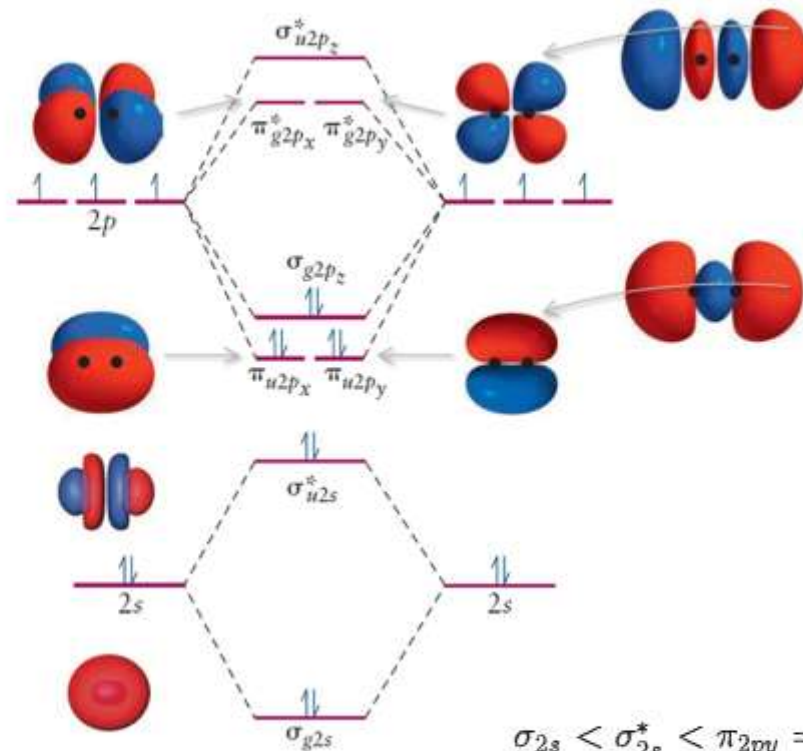
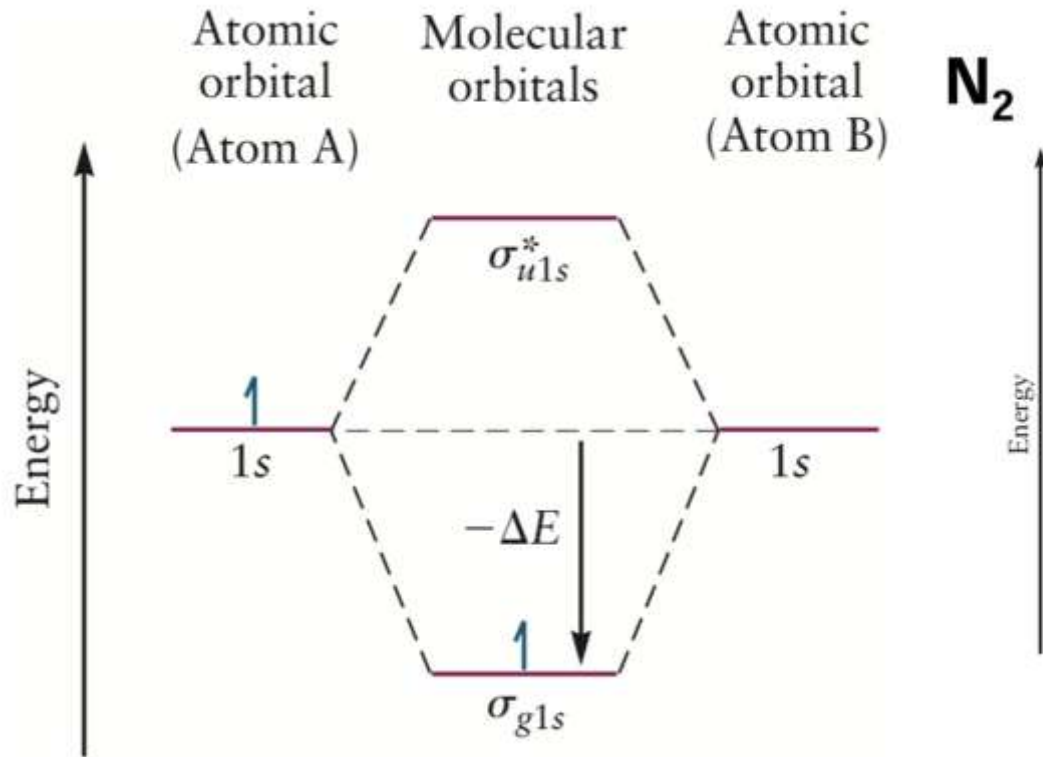
② Compatible orbital symmetries



③ Approximate atomic orbital energies



MO and molecular structure



The phenomenon of s-p mixing occurs when molecular orbitals of the **same symmetry** formed from the **combination of 2s and 2p atomic orbitals** are close enough in energy to further interact, which can lead to a change in the expected order of orbital energies.

$$\sigma_{2s} < \sigma_{2s}^* < \pi_{2py} = \pi_{2pz} < \sigma_{2pz} < \pi_{2py}^* = \pi_{2pz}^* < \sigma_{2pz}^*$$

- Construct all molecular orbitals.
- Half bonding orbitals and half anti-bonding orbitals (what will happen if orbital number is odd?)
- Identify all valence electrons of the molecule.
- Populate the electrons in order of increasing energy (taking care to adhere to Hund's rule).

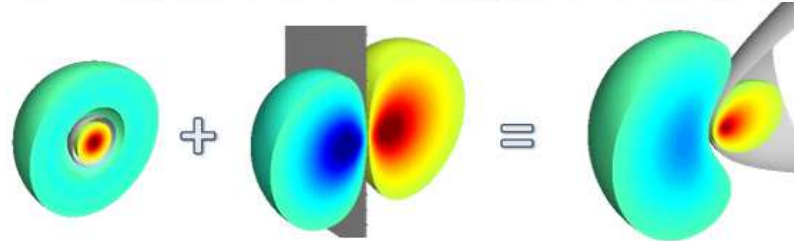
	O_2	O_2^+	NO	NO^+	N_2	CO	CN^-	C_2^{2-}
Electronic configuration	$\sigma^2 \pi^4 \pi^{*2}$	$\sigma^2 \pi^4 \pi^{*1}$	$\sigma^2 \pi^4 \pi^{*1}$	$\pi^4 \sigma^2$	$\pi^4 \sigma^2$	$\pi^4 \sigma^2$	$\pi^4 \sigma^2$	$\pi^4 \sigma^2$
Bond order	2	2.5	2.5	3	3	3	3	3
Bond length (Å)	1.21	1.12	1.15	1.06	1.10	1.13	1.15	1.20
Bond energy (kJ/mol)	494	625	607	?	942	1070	891	839

Hybridization for MO

BeH_2

$$\begin{vmatrix} \varphi_1 \\ \varphi_2 \end{vmatrix} = \begin{vmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{vmatrix} \begin{vmatrix} s \\ p_z \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \begin{vmatrix} s \\ p_z \end{vmatrix}$$

$Be: (1s)^2(\chi_1)^1(\chi_2)^1$



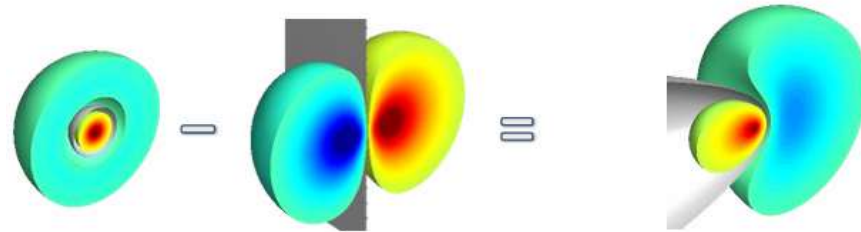
$$\chi_1(r) = \frac{1}{\sqrt{2}} [2s + 2p_z]$$

$$\frac{1}{\sqrt{2}} s$$

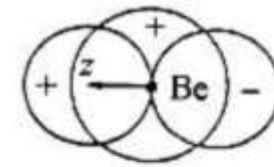
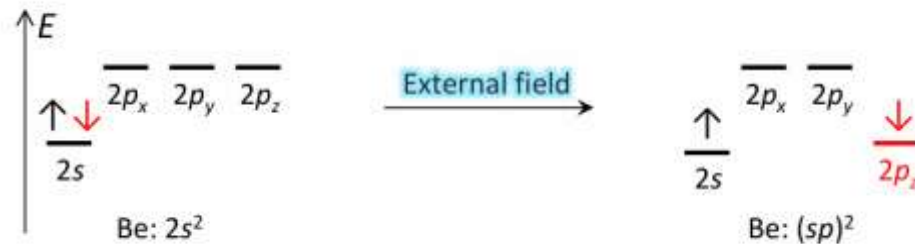
$$\frac{1}{\sqrt{2}} p_z$$

sp hybrid

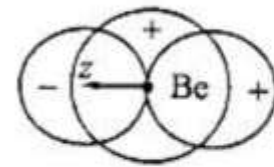
$$\chi_2(r) = \frac{1}{\sqrt{2}} [2s - 2p_z]$$



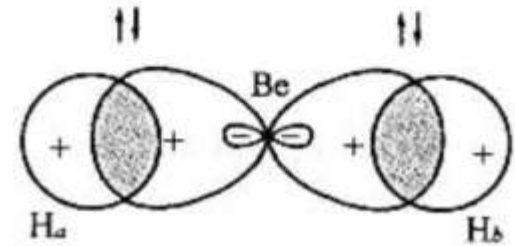
Normalization constant



(a)



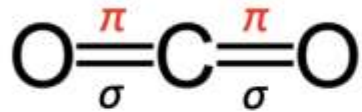
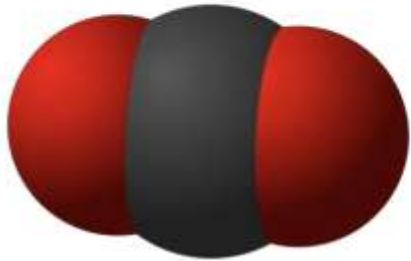
(b)



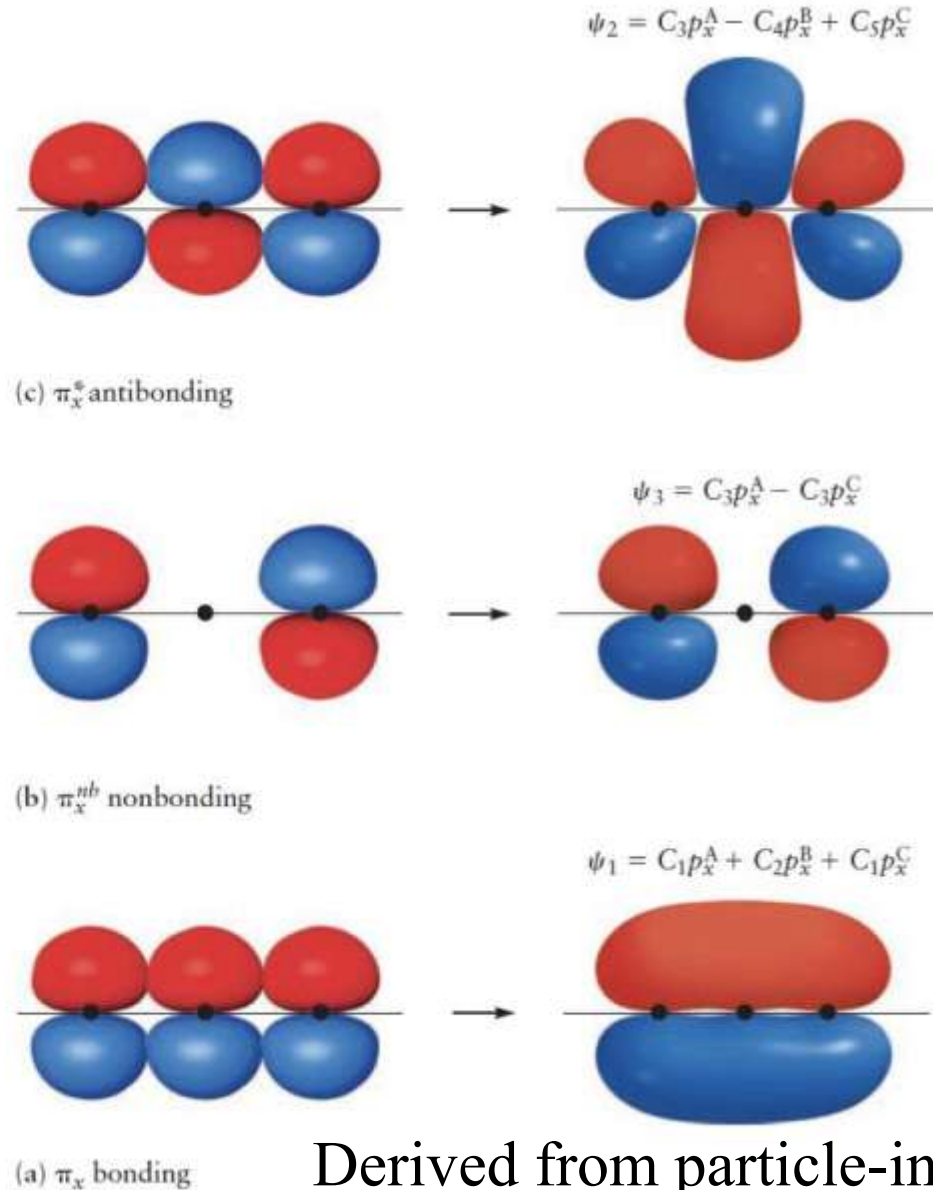
(c)

Hybridization for MO

Linear Triatomic Nonhydrides



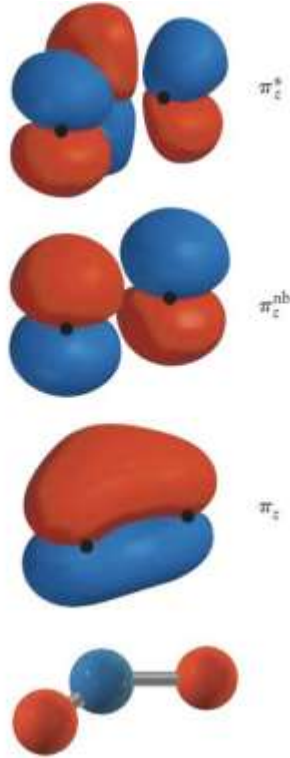
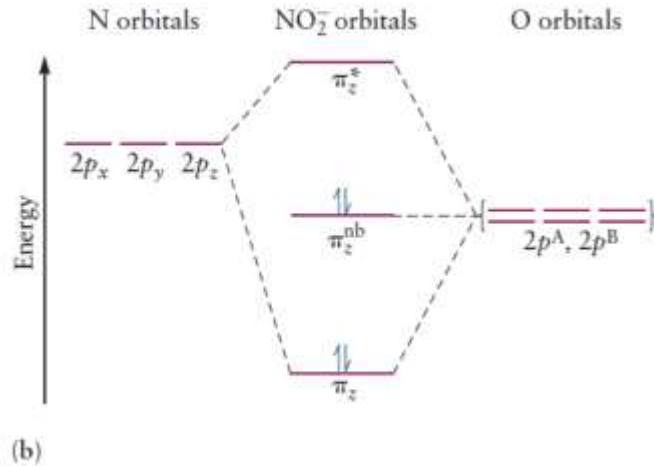
$$E_n = \frac{n^2 h^2}{8mL^2} \quad n = 1, 2, 3, \dots$$



Derived from particle-in-a-box model !

Hybridization for MO

Nonlinear Triatomic Molecules



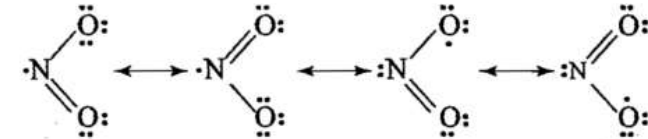
【例 5.5.3】 二氧化氮 NO_2

弯曲形的 NO_2 分子, 键角 134.25° , 键长 119.7 pm 。对它的电子结构, 有三种观点:

(1) 认为单电子处于 N 的一个 sp^2 杂化轨道上, 剩余 p_z 轨道形成 π_3^4 , NO_2 分子的键角远大于 120° , 以及单占杂化轨道容易二聚成 $\text{O}_2\text{N}-\text{NO}_2$ 可以佐证。

(2) 一对孤对电子占据 N 的一个 sp^2 杂化轨道, π 电子形成 π_3^3 , 因为单占的孤对电子能量较高, π_3^3 比 π_3^4 强, 对分子稳定有利。

(3) 介于上述两者之间, 可用下面共振结构式表示:



分子的电子自旋共振及分子轨道的量子化学计算结果比较支持第一种观点。

MO: Regions in a molecule where an electron occupying that orbital is likely to be found

Exercise

The molecular ion HeH^+ has an equilibrium bond length of 0.774 Å. Draw an electron correlation diagram for this molecule, indicating the occupied MOs. If the lowest energy MO has the form, do you expect C_2 to be larger or smaller than C_1 ?

$$C_1 \psi_{1s}^{\text{H}} + C_2 \psi_{1s}^{\text{He}}$$

Exercise

18. When one electron is added to an oxygen molecule, a superoxide ion (O_2^-) is formed. The addition of two electrons gives a peroxide ion (O_2^{2-}). Removal of an electron from O_2 leads to O_2^+ .
- (a) Construct the correlation diagram for O_2^- .
 - (b) Give the valence electron configuration for each of the following species: O_2^+ , O_2 , O_2^- , O_2^{2-} .
 - (c) Give the bond order of each species.
 - (d) Predict which species are paramagnetic.
 - (e) Predict the order of increasing bond dissociation energy among the species.

Exercise

4.5. (0.6 point) (a) The ionization energy of molecular hydrogen (H_2) is *greater* than that of atomic hydrogen (H), but that of molecular oxygen (O_2) is *lower* than that of atomic oxygen (O). Explain. (*Hint*: Think about the stability of the molecular ion that forms in relation to bonding and antibonding electrons.)

(b) What prediction would you make for the relative ionization energies of atomic and molecular fluorine (F and F_2)?

Exercise

An sp^2 hybrid orbital that lies in the xy -plane has the form

$$\psi_{sp^2} = -\sqrt{1/3} \psi_{2s} - \sqrt{1/6} \psi_{2px} + \sqrt{1/2} \psi_{2py}$$

- (a) Show that ψ_{sp^2} is normalized, provided that ψ_{2s} , ψ_{2px} , and ψ_{2py} are all normalized.
- (b) Write the explicit form of this hybrid orbital ψ_{sp^2} and find out the ϕ angle such that ψ_{sp^2} is maximized.

(b) According to Table 5.2, OGB8, let $\sigma = r/a_0$

$$\psi_{2s} = \frac{1}{4\sqrt{2\pi}} a_0^{-3/2} (2-\sigma) \exp(-\sigma/2)$$

$$\psi_{2px} = \frac{1}{4\sqrt{2\pi}} a_0^{-3/2} \sigma \exp(-\sigma/2) \sin\theta \cos\phi$$

Exercise

5.3. (1.2 point) The element xenon (Xe) is by no means chemically inert; it forms a number of chemical compounds with electronegative elements such as fluorine and oxygen. The reaction of xenon with varying amounts of fluorine produces XeF_2 and XeF_4 . Subsequent reaction of one or the other of these compounds with water produces (depending on conditions) XeO_3 , XeO_4 , and H_4XeO_6 , as well as mixed compounds such as XeOF_4 . Predict the structures of these six xenon compounds, using the VSEPR theory.

Exercise

Consider the following C–Cl bond energies in kJ/mol: $\text{C}_6\text{H}_5\text{Cl}$ 398; CH_3Cl 327; $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ 300. Explain why the C–Cl bond is stronger in $\text{C}_6\text{H}_5\text{Cl}$ but weaker in $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$, using resonance structures.

Exercise

The molecular ion S_3N_3^- has the cyclic structure. All S—N bonds are equivalent.

(a) Give six equivalent resonance Lewis diagrams for this molecular ion.

(b) Compute the formal charges on all atoms in the molecular ion in each of the six Lewis diagrams.

Exercise

trans-tetrazene (N_4H_4) consists of a chain of four nitrogen atoms with each of the two end atoms bonded to two hydrogen atoms. Use the concepts of steric number and hybridization to predict the overall geometry of the molecule. Give the expected structure of cis-tetrazene.